Samira Samadi

AI evaluation (benchmark validity, measurement), human–ML collaboration, fairness, robustness and causality

Academic & Industry Experience

Faculty — Max Planck Institute for Intelligent Systems 2020 – Present
Leader of independent research group: Human Aspects of Machine Learning *(incl. two parental leaves)*

- Faculty at the Max Planck ETH Center for Learning Systems *Tübingen, Germany*
- Member of the Cluster of Excellence Machine Learning, University of Tübingen

Leadership

- Founded and led the group, set its research agenda, and secured grants funding the equivalent of six PhD positions (∼1 million euro in research grants).
- Graduated four PhD and two MSc students, who went on to leading academic and industry institutions, including the ELLIS Institute Tübingen, Saarland University, ETH and Prior Labs.

Research

- *AI evaluation:* showed that human personality tests fail basic validity checks on LLMs, and proposed principled, AI-specific tests built on psychometric standards (EAAMO 2025, ICML 2026).
- *Human–ML collaboration:* learning to defer and predictors that complement humans; modeled collaboration as a performative system and tested it in a 1,408-participant study (ICML 2022, NeurIPS 2024, arXiv 2024).
- *Fairness and robustness:* causally grounded distributionally robust optimization, individual fairness and collective counterfactual explanations (AAAI 2024, NeurIPS 2024, AAAI 2025, TMLR 2025, NeurIPS 2026).

Research Intern — Microsoft Research, FATE group January – May 2019
Human-in-the-loop facial recognition systems (CHI 2020 Workshop) *New York, USA*
Mentors: Jennifer Wortman Vaughan, Hanna Wallach

Research Intern — Toyota Technological Institute at Chicago May – August 2018
Individual fairness for structured outcomes *Chicago, USA*
Mentor: Avrim Blum

Research Intern — Sentient Technologies May – August 2016
Interactive visual search for online retail, shipped to production *San Francisco, USA*
Mentor: Phil Long

Research Associate — University of Waterloo August 2014 – August 2015
Finding meaningful cluster structure in data with background noise (ALT 2016) *Waterloo, Canada*
Mentor: Shai Ben-David

Education

PhD in Machine Learning — Georgia Institute of Technology 2015 – 2020
Thesis: Human Aspects of Machine Learning *Atlanta, USA*
Advisor: Santosh Vempala; College of Computing Dissertation Award, honorable mention

MSc in Machine Learning — University of British Columbia 2012 – 2014
Thesis: Near-Optimal Herding (COLT 2014) *Vancouver, Canada*
Advisor: Nick Harvey

BSc in Pure Mathematics — Sharif University of Technology 2007 – 2011
Iranian Mathematical Olympiad semi-finalist in high school (top ∼1,000, twice) *Tehran, Iran*

Selected Publications (for the full list, see Google Scholar)

Position: Stop evaluating AI with human tests, develop principled, AI-specific tests instead, ICML, 2026.
Collective counterfactual explanations: Balancing individual goals and collective dynamics, NeurIPS, 2026.
Challenging the validity of personality tests for large language models, EAAMO, 2025.
Designing ambiguity sets for DRO using structural causal optimal transport, AAAI, 2025.
Bridging causality, individual fairness, and adversarial robustness in the absence of structural causal model, TMLR, 2025.
Wasserstein DRO through the lens of structural causal models and individual fairness, NeurIPS, 2024.
A unifying post-processing framework for multi-objective learn-to-defer problems, NeurIPS, 2024.
Causal adversarial perturbations for individual fairness and robustness in heterogeneous data spaces, AAAI, 2024.
A dynamic model of performative human-ML collaboration: Theory and empirical evidence, preprint, 2024.